Equilibrium and missing forces

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a missing force when acceleration is zero.

SUCCESS CRITERIA - I can...

1. Recognize rest and constant velocity as possible equilibrium.
2. Find missing forces when the net force is zero.
3. Explain why turning is not equilibrium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equilibrium” and “balanced” to explain your answer. Sentence starters are optional.

equilibrium: zero net force and zero acceleration

balanced: adding to zero net force in this force context

tension: the pulling force through a stretched rope or cable

01 Warm-up: bring back an earlier idea

A **3 kg** ball is moving upward at **5 m/s** after leaving a hand. Ignore air resistance; use g=**10 N/kg**.
1. **Draw its free-body diagram**.
2. **Explain why each force belongs** in it.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

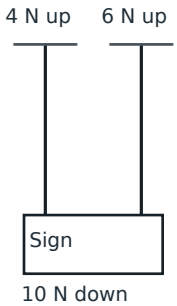
02 Worked example: explain the prediction

Reference: $\Sigma F_x = 0$; $\Sigma F_y = 0 \Rightarrow a = 0$.
Equilibrium allows rest or constant velocity. Resolve missing forces along each axis. Use g = **9.8 N/kg**.

Worked reasoning: A sign has weight **10 N** down and two vertical upward pulls of **4** and **6 N**. The vector sum is zero, so equilibrium is possible.

Your turn: 1. Draw and **label all three forces**.
2. **Name the equation** that checks vertical balance.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

03 Find the balance

The box moves right at steady velocity. A rope pulls **20 N** right. **Find friction** and net force.

Sentence starter: I found … [unit]. This value means … in this situation.



The dot represents the object. Steady rightward velocity

Marzano check - circle: ① ② ③

04 A still lamp

A lamp weighs **30 N** and hangs from one vertical rope without accelerating. **Add both forces** to the diagram and **find the rope tension**.

Sentence starter: I found … [unit]. This value means … in this situation.



Add and label force arrows.

Stationary lamp: weight 30 N.

Marzano check - circle: ① ② ③

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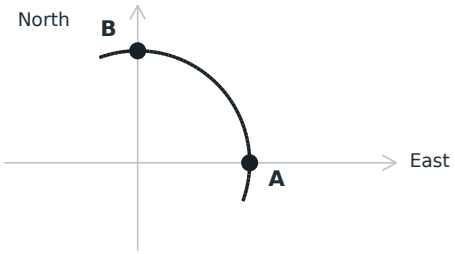
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Parked or cruising?

Compare a parked car, a car moving straight at steady speed, and a car turning at steady speed. Which can have balanced forces? On the curve, add velocity arrows at A and B to **explain the turning case**.

Sentence starter: Both … . They differ in … because … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A goes **7 m** *east* and **5 m** *north*. B then goes **4 m** *west* and **9 m** *south*. **Draw the route** and **find the endpoint relative** to the start.
Draw the component vectors and resultant; **label lengths**.

Word help: component = the part of a vector along one chosen axis

Sentence starter: I found … [unit]. This value means … in this situation.

North / +y (m)



East / +x (m)

Marzano check - circle: ① ② ③

07 Exit check

A box at rest has a **12 N** force right and a **5 N** force left.
1. **Add the missing horizontal force** needed for equilibrium.
2. **Write an equation** that checks the balance.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice … . This supports … because … .



A third horizontal force is missing.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. **Find three different pairs of** nonnegative upward pulls that balance the same **10 N** sign.
2. **Explain what stays constant**.

Sentence starter: Both … . They differ in … because … .

Case	Upward pull A (N)	Upward pull B (N)
1	____	____
2	____	____
3	____	____

Marzano check - circle: ① ② ③